Agentic Identity: Three Architectural Pathways

An [Un]Prompted Lighting talk

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The Identity Crisis

The Problem: Identity systems were designed for humans, not autonomous AI.

The Shift: Agents are decision-making workloads, not just users or service accounts.

The Question: How do we assign identity, trust, and accountability to autonomous software?

Three Architectural Pathways

1. Directory-Based

Agents as Digital Employees

2. Workload-Native

Agents as Infrastructure Workloads

3. Sovereign

Agents as Independent Actors

Pathway 1: Directory-Based Identity

```
// Analogy: Agents as Digital Employees
const directoryIdentity = {
   mechanism: "Enterprise IAM (Entra ID, Okta)",
   authentication: [ "OAuth 2.0 Client Credentials", "X.509
   Certificates" ],
   authorization: [ "RBAC", "ABAC" ]
};
```

Pathway 1: Pros & Cons

Strengths

+ Centralized Governance
+ Auditability
+ Mature Tooling

Weaknesses

– Long-Lived Credentials (Theft Risk)
– Poor for Ephemeral Agents
– Cross-Org Complexity

Best Fit: Enterprise AI, Internal Automation

PATHWAY 2: WORKLOAD-NATIVE

"Agents as Infrastructure Workloads"

MECHANISM

Identity is issued by the runtime environment itself.

```
> SPIFFE / SPIRE
> Kubernetes
```

IDENTITY

Short-lived, automatically rotated SVIDs.

type: X.509 | JWT

ttl: 60m

TRUST

Established via workload attestation and mTLS.

```
verify: node_attest
verify: workload_hash
```

Pathway 2: Pros & Cons

Analysis of Workload-Native Identity Model

Strengths

Zero Standing Credentials

Highly Automated & Secure

Built for Microservices

Weaknesses

Complex Infrastructure

Disconnected from Governance (without tying to OPA or Cedar

Poor Cross-Org Identity

>> BEST_FIT: DevOps-heavy orgs, large-scale AI orchestration

Pathway 3: Sovereign Identity

"Agents as Independent Actors"

Self-Owned Identity

Identity exists independent of any organization using Decentralized Identifiers (DIDs).

Verifiable Credentials

Agents carry digital credentials issued by trust authorities to prove capabilities.

Decentralized Trust

Authentication via cryptographic challenge/response, not centralized directories.

identity_config.jso
n

```
"agent_id": "did:key:z6Mkf ... ",
"auth_method": "cryptographic_challenge",
"credentials": [
{
"type": "VerifiableCredential",
"issuer": "did:web:trust-authority.org",
"claim": "AuthorizedTrader"
}
]

// No central directory required
// Portable across organizations
```

Pathway 3: Protocol Analysis

Evaluation of Sovereign Agent Identity Model

Capabilities

- Cross-Org & Decentralized Trust
- Portable Credentials
- Enables Open Agent Economies

Limitations

- Immature Ecosystem
- Unclear Governance Models
- Performance Overhead

>> TARGET_ENV: Agent Marketplaces, Autonomous Commerce

Architectural Comparison

MODEL	IDENTITY SOURCE	TRUST ANCHOR	IDEAL USE CASE
Directory	IAM Provider	Enterprise Directory	Internal AI Agents
Workload	Infrastructure Runtime	Attestation System	Cloud-Native Agents
Sovereign	Cryptographic Identity	Decentralized Trust	Open Agent Ecosystems

The Future of Agent Identity

> Legacy: "Who is the user?"
> Agentic Era: 3 New Questions

Who created/ tasked this agent?
// ATTRIBUTION
01

What code/ action is it running?
// INTEGRITY
02

What authority does it act under?
// DELEGATION
03

[CRITICAL_WARNING]

"If you don’t give your agents a secure identity, they will borrow yours, and the audit logs may not be kind."

Connect with me

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For everyone else: